Sexism Detection in English Texts by Classification Model

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Introduction

Natural Language Processing (NLP) plays an increasingly important role in the moderation of online content. Among the various challenges in this domain, detecting harmful or biased language such as sexism is particularly important for improving the safety and inclusivity of online platforms.

This project focuses on the task of **sexism detection in English texts**, formulated as a binary classification problem. Given a text, the objective is to determine whether it contains sexist content or not.

To address this problem, we first explore a **classical machine learning approach** based on Naïve Bayes combined with different text representations such as Bag-of-Words, TF-IDF, and n-grams. These models are simple, interpretable, and provide a strong baseline for text classification tasks.

In a second step, we extend the study by using a **pretrained transformer model (BERTweet)**, which leverages contextual embeddings to better capture semantic and linguistic nuances in text.

Finally, beyond classification performance, this project also aims to provide a **linguistic and interpretability analysis**, as well as a visualization of text embeddings, in order to better understand the structure of the data and the behavior of the models.

Dataset

The dataset used in this project is composed of English texts labeled as either **sexist** or **non-sexist**. It is split into training and test sets.

It can be found on the kaggle website at the page **Sexism Detection in English Text** **.**

The training set contains approximately **14,000 samples**, among which only **3,398 are labeled as sexist**. This indicates a clear **class imbalance**, with sexist texts representing a minority of the data.

This imbalance has an important impact on model performance, as classifiers may tend to favor the majority class (non-sexist), leading to lower recall on the sexist class. Therefore, particular attention is given to evaluation metrics such as **recall and F1-score** for the sexist category.

The models are tested on a separate file containing **4,000 samples**.

Methodology

This project explores two complementary approaches for sexism detection: a **classical machine learning pipeline based on Naïve Bayes**, and a **transformer-based approach using a pretrained BERTweet model**.

1. Naïve Bayes Approach

The first part of the project focuses on interpretable models using traditional NLP techniques.

Preprocessing

Several preprocessing steps were applied to the text data:

- lowercasing
- punctuation removal
- tokenization
- stopword removal
- optional stemming

As part of this step, a **custom stopword list** was introduced instead of relying solely on standard English stopwords.

The objective was to preserve potentially informative words, especially **gender-related pronouns** such as *"she"*, *"her"*, or *"women"*, which may carry important signals for sexism detection. In standard stopword lists, such terms are often removed, even though they can be highly relevant in this context.

This highlights that stopword removal should be **task-dependent**, rather than applied systematically.

Feature Extraction

Different text representations were explored:

- **Bag-of-Words (CountVectorizer)**
- **TF-IDF (TfidfVectorizer)**
- **n-grams**, including **bigrams (1,2)**

Particular attention was given to controlling feature sparsity when using n-grams. A minimum document frequency parameter (min_df) was introduced to remove rare and noisy features.

Classification

The main classifier used was:

- **Multinomial Naïve Bayes**

Several configurations were evaluated to compare the impact of preprocessing, feature representation, and n-gram usage.

2. Transformer-based Approach (BERTweet)

In addition to classical methods, a pretrained transformer model was used:

- **Model**: BERTweet (NLP-LTU/bertweet-large-sexism-detector)
- **Architecture**: RoBERTa-based model trained on social media data

Unlike Bag-of-Words models, BERTweet relies on **contextual embeddings**, meaning that each word representation depends on its surrounding context.

This allows the model to capture:

- semantic meaning
- word order
- subtle linguistic patterns

The model was used directly for classification without additional fine-tuning.

3. Embedding Analysis and Visualization

To better understand how texts are represented in high-dimensional space, embeddings extracted from BERTweet were analyzed.

A dimensionality reduction pipeline was applied:

- **PCA** to reduce noise and dimensionality
- **t-SNE** to project embeddings into 2D space

This visualization enables qualitative analysis of the embedding structure and helps assess whether the model separates sexist and non-sexist texts in a meaningful way.

Overall, this methodology allows a comparison between **interpretable classical models** and **contextual deep learning representations**, while also providing insights into the linguistic and semantic structure of the data.

Results

Model	**Accuracy**	**Precison (sexist)**	**Recall (sexist)**	**F1 score**
Raw + CountVectorizer + MultinomialNB	0.8093	0.7018	0.3711	0.4855
Cleaned + CountVectorizer + MultinomialNB	**0.8153**	0.6848	0.4412	0.5367
Cleaned + TF-IDF + MultinomialNB	0.77	0.9630	0.0536	0.1016
Raw + Bigram, min_df = 3 + MultinomialNB	0.8040	0.6099	**0.5320**	**0.5683**
Cleaned + Bigram, min_df = 3 + MultinomialNB	0.8060	0.6174	0.5258	0.5679
BERTweet (pretrained)	**0.8923**	**0.7758**	**0.7814**	**0.7786**

Several models were evaluated to compare different text representations and classification approaches for sexism detection.

The Naïve Bayes baseline using raw text and CountVectorizer achieved an accuracy of 0.8093, but showed limited performance on the sexist class, with a recall of 0.3711. Introducing preprocessing improved the results, with the cleaned CountVectorizer

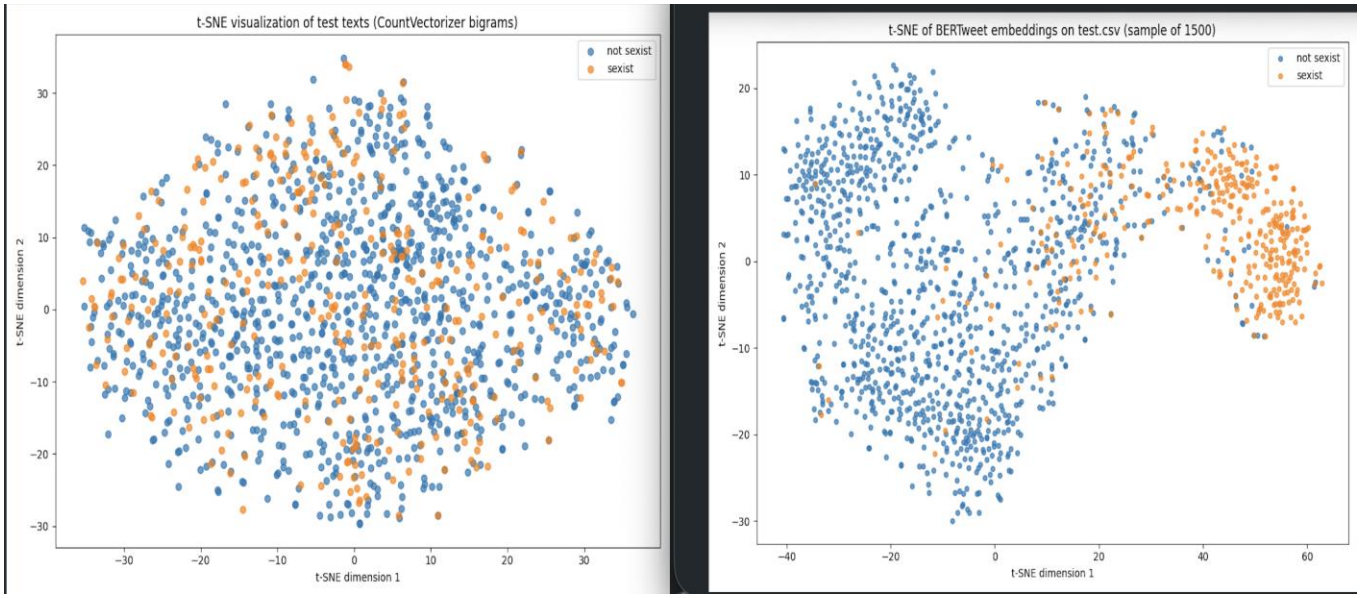
model reaching the best overall accuracy (0.8153) and a higher F1-score on the sexist class (0.5367).

The TF-IDF representation performed significantly worse. Although it achieved a very high precision (0.9630), its recall dropped to 0.0536, indicating that the model failed to identify most sexist texts. This confirms that TF-IDF is not well suited for this task when combined with Naïve Bayes.

The introduction of bigrams led to further improvements when combined with feature filtering. In particular, using CountVectorizer with n-grams (1,2) and a minimum document frequency (min_df=3) significantly improved performance on the sexist class. The best configuration achieved a recall of 0.5320 and an F1-score of 0.5683, making it the most effective model for detecting sexist content.

Finally, the transformer-based approach using BERTweet was evaluated. In addition to strong classification performance, the embedding visualization using t-SNE revealed a clear separation between sexist and non-sexist texts, indicating that the model captures meaningful semantic structures.

Embedding Projector (t-SNE)



To better understand how texts are represented in embedding space, we project high-dimensional representations into two dimensions using PCA followed by t-SNE.

The first visualization corresponds to embeddings obtained from classical representations (e.g., Bag-of-Words / TF-IDF), while the second one uses embeddings extracted from the BERTweet model.

The comparison between the two plots highlights a clear difference in structure. In the case of classical representations, the projection shows a more scattered distribution with limited separation between the classes. This indicates that these representations do not capture strong semantic relationships between texts.

In contrast, the BERTweet-based visualization exhibits a much clearer separation between sexist and non-sexist texts. A dense cluster of sexist examples emerges, while non-sexist texts appear more dispersed. This suggests that the model captures meaningful semantic similarities between texts of the same class.

Additionally, a mixed region can be observed, corresponding to more ambiguous or subtle cases, which are typically more difficult to classify.

Overall, this comparison confirms that contextual embeddings produced by transformer models provide a more structured and informative representation space than classical count-based approaches.

These observations naturally lead to a broader discussion of the strengths and limitations of the different approaches.

Discussion

The results highlight several important aspects of text classification for sexism detection.

First, count-based representations proved to be more effective than TF-IDF when used with Naïve Bayes. This suggests that raw frequency information is more informative than normalized weighting for this specific task. In contrast, TF-IDF overly penalizes frequent words, which may still carry important signals in sexist language.

Second, preprocessing plays a different role depending on the representation. While it improves unigram-based models by reducing noise and normalizing text, it tends to degrade performance for bigram models. This is likely because aggressive preprocessing alters the natural structure of phrases, which are crucial for capturing meaningful patterns in n-grams.

Third, the use of bigrams significantly improves the detection of sexist texts when combined with appropriate filtering. The parameter min_df=3 was particularly important, as it removes rare and noisy features, reducing sparsity and improving generalization. This demonstrates that the effectiveness of n-gram models strongly depends on controlling feature explosion.

Finally, the comparison with BERTweet highlights the limitations of classical models. While Naïve Bayes offers interpretability and simplicity, transformer-based models provide richer semantic representations. The t-SNE visualization clearly shows that BERTweet embeddings naturally separate the two classes, indicating a strong ability to capture contextual meaning.

Overall, this project illustrates a trade-off between interpretability and performance: simpler models are easier to analyze, while more advanced models capture deeper linguistic patterns.

Conclusion

In this project, we investigated the task of sexism detection in English texts using both classical and modern NLP approaches.

The Naïve Bayes models provided a strong and interpretable baseline. In particular, count-based representations combined with bigrams and appropriate feature filtering proved effective for capturing relevant patterns, especially for the detection of sexist texts. The experiments also highlighted the importance of preprocessing choices and the impact of class imbalance on model performance.

The transformer-based approach using BERTweet demonstrated the advantages of contextual embeddings. The visualization of embeddings through t-SNE revealed a clear separation between sexist and non-sexist texts, confirming that the model captures meaningful semantic structure.

Overall, the results show a trade-off between **interpretability and performance**: while Naïve Bayes models are easier to analyze and explain, transformer-based models offer superior representational power.

Future work could include fine-tuning the transformer model on the dataset, exploring other classification algorithms, or improving interpretability methods for deep learning models.

This project demonstrates how combining classical NLP techniques, modern architectures, and interpretability tools can provide a comprehensive understanding of a text classification problem.